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General Definitions

Classical partition function:

$$Z = \frac{1}{N!h^{dN}} \int e^{-\beta H(p,q)} d^{dN}p d^{dN}q$$

Separable Hamiltonian: $H = H_p(p) + H_q(q)$

$$Z = \frac{1}{N!h^{dN}} Z_p Z_q$$

$$Z_p = \int e^{-\beta H_p} d^{dN}p, \quad Z_q = \int_V e^{-\beta H_q} d^{dN}q$$

DOS representation:

$$Z(\beta) = \int_0^\infty dE g(E) e^{-\beta E}$$

Factorization: For independent subsystems: $Z_{total} = \prod_i Z_i$

Thermodynamic relations:

$$F = -k_B T \ln Z_N,$$
$$U = -\frac{\partial \ln Z_N}{\partial \beta} = k_B T^2 \frac{\partial \ln Z_N}{\partial T} = -\frac{1}{Z_N} \frac{\partial Z_N}{\partial \beta}$$
$$\ln Z_N = \ln \left(\frac{1}{N!} (Z_1)^N \right) = N \ln Z_1 - \ln N!$$
$$S = k_B \ln Z_N + \frac{U}{T} = k_B (\ln Z_N + \beta U) = \frac{U - F}{T}$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V,N},$$

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T,N},$$

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T,V} = \left(\frac{\partial G}{\partial N} \right)_{T,P}, \quad G = N \mu$$

Extensivity: $S(\lambda E, \lambda V, \lambda N) = \lambda S(E, V, N)$ ($\lambda > 0$)

Ideal Gas (N indistinguishable): $Z_N = \frac{(Z_1)^N}{N!}$,

$$F = -Nk_B T \left[\ln \left(\frac{V}{N \lambda^3} \right) + 1 \right],$$

$$U = Nk_B T^2 \frac{\partial \ln Z_1}{\partial T},$$

$$S = Nk_B \left[\ln \left(\frac{Z_1}{N} \right) + 1 + T \frac{\partial \ln Z_1}{\partial T} \right]$$

Grand Canonical Ensemble

Fundamental Formula

$$\Xi = \prod_j \xi_j, \quad \xi_j = \sum_{n_j \in \mathcal{A}} e^{-\beta n_j (\epsilon_j - \mu)}$$

where $\mathcal{A}$ is the occupation constraint ($\{0, 1\}$ fermions, $\{0, 1, \dots\}$ bosons).

Thermodynamic Relations

$$\langle N \rangle = z \frac{\partial \ln \Xi}{\partial z} = k_B T \frac{\partial \ln \Xi}{\partial \mu}, \quad P = k_B T \frac{\partial \ln \Xi}{\partial V}, \quad S = k_B (\ln \Xi + \beta \langle E - \mu N \rangle)$$

Fermions vs. Bosons

- Fermions:** $\xi_j = 1 + e^{\beta(\mu - \epsilon_j)}$ (Pauli exclusion, $n_j \in \{0, 1\}$)
Half-filling: $\mu = \epsilon_c$ (symmetric spectrum) independent of T
- Bosons:** $\xi_j = (1 - e^{\beta(\mu - \epsilon_j)})^{-1}$ (no limit, $n_j \in \{0, 1, \dots\}$)
Constraint: $\mu < \epsilon_{\min}$

Non-interacting Systems

Classical: $\Xi = e^{zZ_1}$, $\langle N \rangle = zZ_1$

Ideal gas: $\Xi = \exp(zV/\lambda^3)$, $\langle N \rangle = zV/\lambda^3$, $PV = \langle N \rangle k_B T$

Adsorption (Langmuir): $\Xi = (1 + ze^{\beta \epsilon_0})^M$,

$$\theta = \frac{ze^{\beta \epsilon_0}}{1 + ze^{\beta \epsilon_0}} \quad (\text{coverage})$$

Density of States

$$g(E) = \frac{d\Gamma}{dE}, \quad \Omega(E) = g(E) \Delta E$$

Spin Degeneracy g_s :

- Spin-1 particles: $g_s = 3$ (if massive), $g_s = 2$ (if massless)
- General: $g_s = 2s + 1$ for spin- s massive particles

General Formula (with h):

$$\Gamma(\epsilon) = g_s \left(\frac{L}{h} \right)^d \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} \left(\frac{2m\epsilon}{1} \right)^{d/2}$$

$$g(\epsilon) = g_s \frac{V_d}{h^d} \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} (2m)^{d/2} \frac{d}{2} \epsilon^{d/2 - 1}$$

General Scaling Law (Master Formula):

Procedure: (1) Identify dispersion $\epsilon(p)$; (2) Use phase space: $\Gamma(\epsilon) = g_s \frac{V}{h^d} \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} [p(\epsilon)]^d$; (3) Differentiate to get $g(\epsilon)$.

For a dispersion $\epsilon \propto p^s$ in d dimensions:

$$g(\epsilon) \propto \epsilon^{\frac{d}{s} - 1}$$

Dimension	Massless $E = pc$	Massive $E = p^2/2m$	$N(\leq n)$		
	$\Gamma(\leq E)$	$g(E)$	$\Gamma(\leq E)$	$g(E)$	$N(\leq n)$
$3D$	E^3	E^2	$E^{3/2}$	$E^{1/2}$	n^3
$2D$	E^2	E	E	E^0	n^2
$1D$	E	E^0	$E^{1/2}$	$E^{-1/2}$	n

Check: $3D$ Electrons ($d = 3, s = 2$) $\rightarrow \epsilon^{1/2}$. Graphene ($d = 2, s = 1$) $\rightarrow \epsilon^1$.

A. Non-Relativistic ($\epsilon = p^2/2m$):

$3D$ Free Electron (Bulk): $\Omega_3 = 4\pi$, $p = \sqrt{2m\epsilon}$, $\frac{dp}{d\epsilon} = \sqrt{\frac{m}{2\epsilon}}$

$$g_{3D}(\epsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{\epsilon}$$

$2D$ Quantum Well: $\Omega_2 = 2\pi$, $\frac{dp}{d\epsilon} = \frac{m}{p} = \sqrt{\frac{m}{2\epsilon}}$

$$g_{2D}(\epsilon) = \frac{Am}{\pi \hbar^2} \quad (\text{constant!})$$

$1D$ Quantum Wire: $\Omega_1 = 2$, $dp/d\epsilon = m/p$

$$g_{1D}(\epsilon) = \frac{L}{\pi \hbar} \sqrt{\frac{m}{2\epsilon}} \quad (\text{Van Hove singularity})$$

B. Massless Bosons ($\epsilon = c_s p$):

$3D$ Photons/Phonons: $p = \epsilon/c$, $dp/d\epsilon = 1/c$

$$g_{3D}(\epsilon) = \frac{g_s V \epsilon^2}{\pi^2 \hbar^3 c^3} \quad (\text{photons: } g_s = 2 \text{ polarizations})$$

$2D$ Graphene (Dirac): $\epsilon = \hbar v_F k$, $p = \hbar k$

$$g_{2D}(\epsilon) = \frac{g_s A \epsilon}{\pi \hbar^2 v_F^2} \quad (\text{graphene: } g_s = 4 \text{ valley+spin})$$

$$g(\epsilon) = \frac{g_{total} A \epsilon}{2\pi (\hbar v)^2}$$

$$E_{n_x, n_y, n_z} = \frac{\hbar^2 \pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$$

C. Relativistic Massive:

$$\epsilon = \sqrt{p^2 c^2 + m^2 c^4} - mc^2$$

$$p = \frac{1}{c} \sqrt{\epsilon^2 + 2mc^2 \epsilon}, \quad \frac{dp}{d\epsilon} = \frac{\epsilon + mc^2}{c^2 p}$$

$$g_{3D}(\epsilon) = \frac{4\pi V}{\hbar^3 c^3} (\epsilon + mc^2) \sqrt{\epsilon^2 + 2mc^2 \epsilon}$$

Limits:

- $\epsilon \gg mc^2$: $g \propto \epsilon^2$ (photon-like)
- $\epsilon \ll mc^2$: $g \propto \sqrt{\epsilon}$ (non-relativistic)

Ideal Gas

Thermal de Broglie:

$$\lambda = \sqrt{\frac{2\pi \hbar^2}{mk_B T}} = \frac{h}{\sqrt{2\pi mk_B T}}$$

Quantum concentration: $n_Q = \lambda^{-3}$

Single particle: $Z_1 = \frac{V}{\lambda^3} = V n_Q$

N particles:

$$Z_N = \begin{cases} (Z_1)^N & \text{distinguishable} \\ \frac{(Z_1)^N}{N!} & \text{indistinguishable} \end{cases}$$

Free energy: $F = -Nk_B T \left[1 + \ln \left(\frac{g_s n_Q}{n} \right) \right]$

Chemical potential:

$$\mu = k_B T \ln \left(\frac{n \lambda^3}{g_s} \right) = k_B T \ln \left(\frac{n}{g_s n_Q} \right)$$

Sackur-Tetrode:

$$S = Nk_B \left[\frac{5}{2} + \ln \left(\frac{g_s n_Q}{n} \right) \right]$$

$$\frac{S_{\text{trans+rot}}}{N} = k_B \left[\ln \left(\frac{V}{N \lambda^3} \frac{T}{\Theta_r} + \frac{7}{2} \right) \right]$$

where $n_Q = \left(\frac{mk_B T}{2\pi \hbar^2} \right)^{3/2}$ $S \sim Nk_B \ln(V T^{f/2}) = \text{const} \implies V T^{f/2} = \text{const}$

d-Dimensional Gas

$$Z_1 = \frac{V_d}{\lambda^d} e^{\beta \epsilon_d}, \quad \Xi = \exp \left(\frac{z g_s V_d}{\lambda^d} \right)$$

Equipartition Theorem

Each quadratic DoF: contributes $\frac{1}{2} k_B T$ to U

Linear DoF: contributes $k_B T$

General: $U = \frac{f}{2} Nk_B T$, $C_V = \frac{f}{2} Nk_B$ (f = quadratic DoF)

Counting DoF: Trans: $f_t = 3$; Rot: $f_r = 0$ (monat.), 2 (linear), 3 (non-lin.); Vib: $f_v = 2(3N_{\text{atoms}} - 5)$ (linear), $2(3N_{\text{atoms}} - 6)$ (non-lin.)

Active modes: Vibrations need $k_B T \sim \hbar \omega_{\text{vib}}$ (frozen at low T)

Rotational partition:

$$Z_{\text{rot}} = \sum_{\ell=0}^\infty (2\ell + 1) \exp \left[-\frac{\ell(\ell+1)\Theta_r}{T} \right], \quad \Theta_r = \frac{\hbar^2}{2Ik_B} \quad (\text{rot. temp.})$$

$$\text{High-}T \text{ limit: } Z_{\text{rot}}^{(1)} = \frac{T}{\Theta_r (2n_{\text{homo}} + 1n_{\text{hetero}})}$$
$$\text{Vibrational: } Z = \frac{e^{-\Theta_v/2T}}{1 - e^{-\Theta_v/T}}, \quad \frac{C_V}{k_B} = \left(\frac{\Theta_v}{T} \right)^2 \frac{e^{\Theta_v/T}}{(e^{\Theta_v/T} - 1)^2}$$

$$\text{Electronic (2-level): } Z = g_0 + g_1 e^{-\Theta_e/T}, \quad \frac{C_V}{k_B} = \left(\frac{\Theta_e}{T} \right)^2 \frac{g_0 g_1 e^{\Theta_e/T}}{(g_0 e^{\Theta_e/T} + g_1)^2}$$

Common cases: Monatomic: $f = 3$, $C_V = \frac{3}{2} Nk_B$; Diatomic (no vib): $f = 5$, $C_V = \frac{5}{2} Nk_B$; Non-linear (no vib): $f = 6$, $C_V = 3Nk_B$

$$\langle V \rangle = \frac{k_B T}{n} \quad (\text{where } n \text{ is the power of the potential})$$

Quantum Statistics

Occupation Numbers

Quantum regime: $n \gtrsim n_Q$ (wavefunctions overlap)

General occupation:

$$\langle n_\epsilon \rangle = \frac{1}{e^{\beta(\epsilon - \mu)} + \eta}$$

$\eta = +1$ (Fermions), $\eta = -1$ (Bosons), $\eta = 0$ (Classical) **Fermions:** μ can be positive, max 1 per state

$$P(n_i) = \frac{e^{-\beta(\epsilon_i - \mu)n_i}}{\Xi_i}$$

$$\langle n_i \rangle = \sum_n n P(n_i) = \frac{\sum_n n e^{-\beta(\epsilon_i - \mu)n}}{\sum_n e^{-\beta(\epsilon_i - \mu)n}}$$

Bosons: $\mu < \epsilon_{\text{ground}}$ required

Total number: $N = \int_0^\infty g(\epsilon) f(\epsilon) d\epsilon$

Grand Potential

$$\Phi = -\eta k_B T \sum_s \ln(1 + \eta e^{\beta(\mu - \epsilon_s)})$$

$$\text{Single state: } Z_{FD} = 1 + e^{-\beta(\epsilon - \mu)}, \quad Z_{BE} = \frac{1}{1 - e^{-\beta(\epsilon - \mu)}}$$

Fermi Gas

$$\text{Fermi energy (3D): } \epsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}, \mu(0) = \epsilon_F$$

$$\text{General (d-dim): } \epsilon_F = \frac{\hbar^2}{2m} \left[\frac{n}{g_s} (4\pi)^{d/2} \Gamma(d/2 + 1) \right]^{\frac{2}{d}}$$

Fermi temperature: $T_F = \epsilon_F / k_B$

$$T = 0 \text{ occupation: } f_{FD}(\epsilon) = \begin{cases} 1 & \epsilon < \epsilon_F \\ 0 & \epsilon > \epsilon_F \end{cases}$$

Ground state energy: $U_0 = \frac{3}{5} N \epsilon_F$

Degeneracy pressure: $P_0 = \frac{2}{5} n \epsilon_F = \frac{2U_0}{3V}$

Sommerfeld ($T \ll T_F$):

$$\mu(T) \approx \varepsilon_F \left[1 - \frac{\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right],$$
$$\mu(T) \approx E_F \left[1 - \frac{\pi^2}{6} \left(\frac{d}{2} - 1 \right) \left(\frac{k_B T}{E_F} \right)^2 \right]$$
$$U \approx U_0 \left[1 + \frac{5\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right], \left[C_V = \frac{\pi^2}{2} N k_B \frac{T}{T_F} \right] \text{ (linear in } T)$$
Relativistic Fermi energy:
$$\varepsilon_F = \sqrt{\hbar^2 c^2 (3\pi^2 n)^{2/3} + m^2 c^4} - m c^2$$

Bose-Einstein Condensation

Total particles:

$$N = \int_0^\infty g(\varepsilon) f_{BE}(\varepsilon) d\varepsilon$$

Critical Temperature: At T_c , excited states reach maximum capacity:

$$N = N_{\text{excited}}^{\text{max}}(T_c)$$
with
$$\mu \rightarrow 0$$

Below T_c : excess particles condense into ground state ($N_0 = N - N_{ex}$) **General T_c derivation:** BEC occurs when excited states saturate. **Condition:** $\mu(T_c) \rightarrow \varepsilon_{min}$. **Equation:** Set $N = N_{excited}(\mu = \varepsilon_{min})$.

$$N_{ex} = \int_{\varepsilon_{min}}^\infty \frac{g(\varepsilon)}{e^{\beta(\varepsilon - \varepsilon_{min})} - 1} d\varepsilon \tag{1}$$

Procedure: 1. Find DOS $g(\varepsilon)$. 2. Identify ε_{min} . 3. Sub $x = \beta_c(\varepsilon - \varepsilon_{min})$ to extract T_c . (Usually yields Riemann Zeta $\zeta(n)$ or Gamma functions). Using:

$$\int_0^\infty \frac{\varepsilon^{s-1}}{e^{\varepsilon/k_B T} - 1} d\varepsilon = \Gamma(s) \zeta(s) (k_B T)^s$$

$$T_c = \frac{2\pi\hbar^2}{mk_B} \left(\frac{n}{g_s \zeta(d/2)} \right)^{2/d}$$

($\Gamma(3/2) = \sqrt{\pi}/2$, $\zeta(3/2) \approx 2.612$, $n = N/V$)

Condensate fraction:

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2} \quad (T < T_c)$$

Excited particles:

$$N_{ex} = N \left(\frac{T}{T_c} \right)^{3/2}$$

Below T_c :

$$\mu = 0, \quad C_V \propto T^{3/2}, \quad P \propto T^{5/2}$$

Heat capacity jump:

$$\Delta C_V(T_c) \approx 3.66 N k_B$$

Key Points:

- T_c increases with density n , decreases with mass m
- BEC requires $d > 2$ for massive particles
- 2D: No BEC (DOS const $\Rightarrow$ integral diverges)
- 2D with gap E_0 : BEC possible

Magnetization:

$$M = -\frac{\partial F}{\partial B} = k_B T \frac{\partial \ln Z}{\partial B} = \frac{\sum M_i e^{-\beta E_i}}{Z}$$

Susceptibility:

$$\chi = \left. \frac{\partial(M/V)}{\partial B} \right|_{B=0}$$

Massless Bosons

Photons (2 polarizations): $\varepsilon = \hbar \omega = pc$, $\mu = 0$, Bose-Einstein distribution

$$Z_1 = \frac{8\pi V (k_B T)^3}{h^3 c^3}, \quad u(\omega)_{\text{photon}} = \frac{\hbar \omega^3}{\pi^2 c^3} \frac{1}{e^{\beta \hbar \omega} - 1}$$

$$U_{\text{photon}} = \frac{\pi^2 k_B^4 T^4 V}{15 \hbar^3 c^3} \text{ (Stefan-Boltzmann)}$$

Phonons (3 polarizations, Debye cutoff): $\varepsilon = \hbar \omega_s = \hbar c_s |k|$, sound velocity c_s

$$\omega_D = c_s \left(\frac{6\pi^2 N}{V} \right)^{1/3} \text{ (Debye cutoff)}$$

Constraint: Phonons have $\mu = 0$, require DOS cutoff at ω_D ; photons unlimited but rare at high ω .

Blackbody Radiation

Photon gas: $\mu = 0$ (no number conservation)

Planck:

$$\langle n_\omega \rangle = \frac{1}{e^{\beta \hbar \omega} - 1}$$
$$u(\omega) = \frac{\hbar \omega^3}{\pi^2 c^3} \frac{1}{e^{\beta \hbar \omega} - 1}$$

Stefan-Boltzmann: $U = \frac{\pi^2 k_B^4}{15 \hbar^3 c^3} T^4 V$, $P = \frac{U}{3V}$

Wien's law: $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m}\cdot\text{K}$

Massive Relativistic Gas

Dispersion: $\varepsilon = \sqrt{p^2 c^2 + m^2 c^4}$ Single part. partition function: $Z_1 = \frac{4\pi V m^2 c}{\beta \hbar^3} K_2(\beta m c^2)$ **Limits:** *Ultra-rel* ($\beta m c^2 \ll 1$): $Z_1 \approx 8\pi V (\frac{k_B T}{\hbar c})^3$ *Non-rel* ($\beta m c^2 \gg 1$): $Z_1 \approx V (\frac{2\pi m k_B T}{\hbar^2})^{3/2} e^{-\beta m c^2}$

Standard Models

Thermodynamics

First Law: $dU = T dS - P dV + \mu dN$,

$$C_V = T \left(\frac{\partial S}{\partial T} \right)_V, \quad C_P = T \left(\frac{\partial S}{\partial T} \right)_P$$

Maxwell Relations & Potentials

Differentials:

$$H = U + PV, \quad F = U - TS, \quad G = U + PV - TS, \quad \Phi = U - TS - \mu N = -PV$$

$$dF = -SdT - PdV + \mu dN$$

$$dG = -SdT + VdP + \mu dN$$

$$d\Phi = -SdT - PdV - Nd\mu$$

Maxwell Relations:

$$\begin{aligned} (\partial T / \partial V)_S &= -(\partial P / \partial S)_V \\ (\partial T / \partial P)_S &= (\partial V / \partial S)_P \\ (\partial S / \partial V)_T &= (\partial P / \partial T)_V \\ (\partial S / \partial P)_T &= -(\partial V / \partial T)_P \end{aligned}$$

Adiabatic Relations

Adiabatic Process ($Q = 0, dS = 0$):

$$\left[\frac{TV^\gamma - 1}{\gamma} = \text{const} \right] \quad \left[\frac{PV^\gamma}{\gamma} = \text{const} \right] \quad \left[\frac{TP^{-\frac{1}{\gamma}}}{\gamma} = \text{const} \right]$$
where $\gamma = C_P/C_V = (f+2)/f$ (f = active DoF).

Common values: Monatomic ($f = 3$): $\gamma = 5/3$; Diatomic ($f = 5$): $\gamma = 7/5$; Non-linear ($f = 6$):

$$\gamma = 4/3.$$

Harmonic Oscillator

Classical: $Z_1 = \frac{k_B T}{\hbar \omega}$, $U = k_B T$, $C_V = k_B$

Quantum: $Z_1 = \frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} = \frac{1}{2 \sinh(\beta \hbar \omega / 2)}$

$$\langle n \rangle = \frac{1}{e^{\beta \hbar \omega} - 1}, \quad U = \hbar \omega \left(\frac{1}{2} + \langle n \rangle \right)$$

$$C_V = k_B \frac{(\beta \hbar \omega)^2 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

Limits: $T \gg \hbar \omega / k_B \Rightarrow U \rightarrow k_B T$, $C_V \rightarrow k_B$; $T \ll \hbar \omega / k_B \Rightarrow U \rightarrow \hbar \omega / 2$, $C_V \rightarrow 0$

Two-Level System (Schottky)

$$Z = 1 + e^{-\beta \varepsilon} \text{ (or } Z = e^{\beta \varepsilon / 2} + e^{-\beta \varepsilon / 2}), \quad U = \frac{\varepsilon}{1 + e^{\beta \varepsilon}},$$

$$C_V = k_B \left(\frac{\beta \varepsilon}{2} \right)^2 \text{sech}^2 \left(\frac{\beta \varepsilon}{2} \right), \text{ peak at } k_B T \approx 0.42 \varepsilon$$

Real Gases (Van der Waals)

Equation: $(P + \frac{aN^2}{V^2})(V - Nb) = Nk_B T$ Virial Expansion: $\frac{P}{k_B T} = n + B_2(T)n^2 + B_3(T)n^3 + \dots$

Second virial coefficient: $B_2(T) = b - \frac{a}{k_B T}$ (*attractive interactions:* $a > 0$, repulsive: $b > 0$ hard-sphere volume)

Critical Point: $V_c = 3Nb$, $P_c = \frac{a}{27b^2}$, $k_B T_c = \frac{8a}{27b}$

Einstein Solid

$\Omega(N, q) = \binom{N+q-1}{q}$
High T : $U \approx Nk_B T$, $C_V \approx Nk_B$
Low T : $C_V \propto e^{-\hbar \omega / k_B T}$ (freeze-out)

Debye Model

$$\Theta_D = \hbar \omega_D / k_B$$

Low T :
$$C_V = \frac{12\pi^4}{5} N k_B \left(\frac{T}{\Theta_D} \right)^3$$
High T : $C_V \rightarrow 3Nk_B$ (Dulong-Petit)

Maxwell-Boltzmann Velocity

$$P(v)dv = 4\pi n \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T} dv$$

$$v_{mp} = \sqrt{\frac{2k_B T}{m}}, \quad \langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}, \quad v_{rms} = \sqrt{\frac{3k_B T}{m}}$$

Ising Model

$$M^2 = \sum_{i=j} s_i^2 + \sum_{i \neq j} s_i s_j, \quad \sum_{i \neq j} s_i s_j = 2 \sum_{\text{pairs}} s_i s_j$$

$$\langle s_i \rangle = \sum_s p(s) s = \sum_i \frac{e^{-\beta \varepsilon_i}}{Z} s$$

$$H = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i \quad (s_i = \pm 1)$$

$$Z = \sum M(M) e^{-\beta U(M)}$$

Mean Field: $m = \tanh(\beta(zJm + h))$

Susceptibility (Mean Field): From $m = \tanh(\dots)$, differentiate w.r.t h :

$$\begin{aligned} \chi &= \text{sech}^2(\beta(zJm + h)) \cdot \beta(zJ\chi + 1) \\ &= (1 - m^2) \beta(zJ\chi + 1) \quad (\because \text{sech}^2 = 1 - \tanh^2) \\ \Rightarrow \chi[1 - \beta zJ(1 - m^2)] &= \beta(1 - m^2) \end{aligned}$$

$$\chi(T, h) = \frac{\beta(1 - m^2)}{1 - \beta zJ(1 - m^2)}$$

Critical temp: $k_B T_c = zJ = 2dJ$

1D Free Energy (Exact, $h = 0$): $f = -\frac{1}{\beta} \ln \lambda_+ =$

$$-\frac{1}{\beta} \ln \left[2e^{\beta J} \cosh(\beta J) \right]$$

1D: No phase transition at $T > 0$ (exponential decay of correlations)

2D Onsager: $k_B T_c = \frac{2J}{\ln(1 + \sqrt{2})} \approx 2.269J$ **Ising Phase**

Transition ($T \approx T_c$):

- $h = 0$ (**Spontaneous SB**): χ diverges at T_c . $\chi(T \rightarrow T_c^+) \propto \frac{1}{T - T_c}$, $\chi(T \rightarrow T_c^-) \propto \frac{1}{2(T_c - T)}$. **Amplitude Ratio:** $\chi_+ / \chi_- = 2$. Asymmetry due to mean field.
- $h \neq 0$ (**Explicit SB**): No phase transition (no singularity). External field breaks Z_2 symmetry explicitly $\Rightarrow m \neq 0$ for all T . $\chi(T)$ shows finite peak (Smooth transition) instead of divergence. Peak broadens/shifts as $h \uparrow$.

Critical Exponents

	Definition	MFT	2D Ising (Onsager)
α	$C \sim t ^{-\alpha}$	0	0
β	$m \sim (-t)^\beta$	1/2	1/8
γ	$\chi \sim t ^{-\gamma}$	1	7/4
δ	$m \sim h^{1/\delta}$	3	15

Curie-Weiss: $\chi = \frac{C}{T - T_c}$, $\chi_{T < T_c} / \chi_{T > T_c} = 1/2$

Saha: $\frac{n_e n_p}{n_H} = \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} e^{-E_I / k_B T}$

Standard Procedures

Find μ from Z :

$$Z_N \rightarrow F = -k_B T \ln Z_N \rightarrow \mu = \frac{\partial F}{\partial N}$$

$$\Gamma(\varepsilon) = \frac{1}{h^d} \iint_{H \leq \varepsilon} d^d q d^d p \rightarrow g(\varepsilon) = \frac{d\Gamma}{d\varepsilon}$$

$$Z \rightarrow F = -k_B T \ln Z \rightarrow M = -\frac{\partial F}{\partial B}$$

$$\chi = \left. \frac{\partial(M/V)}{\partial B} \right|_{B=0}$$

BEC T_c : Set $N = N_{ex}(T_c, \mu = 0)$

Phase equilibrium:

$$\mu_1(n_1, T) = \mu_2(n_2, T)$$

d-dim sphere volume:

$$V_d(p) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} p^d$$

$$\Gamma(E) = \frac{V}{h^d} \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} (2mE)^{d/2}$$

Gibbs Entropy:

S = -k_B \sum_i p_i \ln p_i

Canonical from S max: Maximize S = -k_B \sum p_i \ln p_i with constraints \sum p_i = 1, \sum p_i E_i = U \Rightarrow p_i = e^{-\beta E_i} / Z (Lagrange multipliers).

Microcanonical:

S = k_B \ln \Omega, \quad \frac{1}{T} = \frac{\partial S}{\partial U}, \quad \Omega = \binom{N}{k}

Anharmonic oscillator (g^4): Z_x \approx Z_{HO} \left(1 - \frac{3g}{\beta m^2 \omega^4}\right)

Chemical equilibrium procedure:

- 1. Find \mu_i = k_B T \ln(n_i \lambda^{d_i}) - \epsilon_i for each phase
- 2. Set \mu_1 = \mu_2 at equilibrium
- 3. Solve: n_2 = n_1 \lambda^{d_1 - d_2} e^{\beta(\epsilon_2 - \epsilon_1)}
- 4. Check dimensional consistency: [n_d] = L^{-d}

Mean-field/Self-consistent:

- 1. Replace interactions with effective field: H_{eff} = h_{eff} \sum_i s_i
- 2. Compute mean: \langle s \rangle = \tanh(\beta h_{eff})
- 3. Self-consistency: h_{eff} = z J \langle s \rangle + h_{ext}
- 4. Solve: m = \tanh(\beta (z J m + h))

Constants: k_B = 1.381 \times 10^{-23} J/K, \hbar = 1.055 \times 10^{-34} J\cdot s, 1 eV = 1.602 \times 10^{-19} J, 1 J = 6.242 \times 10^{18} eV, k_B T|_{300K} \approx 1/40 eV \approx 0.026 eV

Fluctuations

Canonical (N, V, T): \langle (\Delta E)^2 \rangle = k_B T^2 C_V

Grand Canonical (\mu, V, T): \langle (\Delta N)^2 \rangle = k_B T \left(\frac{\partial \langle N \rangle}{\partial \mu} \right)_{T,V} = \frac{N^2 k_B T}{V} \kappa_T \quad \frac{\sigma_N}{\langle N \rangle} = \frac{1}{\sqrt{N}} \sqrt{\frac{k_B T \kappa_T}{v}}

Relative fluctuations vanish as 1/\sqrt{N}.

Examples

Partition Functions

(3D free particle):

\Gamma(\epsilon) = \frac{V}{h^3} \frac{4\pi}{3} (2m\epsilon)^{3/2}

g(\epsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{\epsilon}

Z_1 = \int_0^\infty g(\epsilon) e^{-\beta \epsilon} d\epsilon = \frac{V}{\lambda^3}, \quad \lambda = \frac{h}{\sqrt{2\pi m k_B T}}

2. Particle in gravity (U = mgz):

Z_1 = \frac{A}{\lambda^2} \int_0^L e^{-\beta mgz} dz = \frac{A}{\lambda^2} \frac{1 - e^{-\beta mgL}}{\beta mg}

High T:

Z_1 \approx \frac{AL}{\lambda^2} = \frac{V}{\lambda^2}

\mu_{tot} = \mu_{int}(n(\vec{r}), T) + V_{ext}(\vec{r}) = \text{const}

3. Harmonic trap (U = \frac{1}{2} m \omega^2 x^2):

Z_q = \int_{-\infty}^\infty e^{-\beta m \omega^2 x^2 / 2} dx = \sqrt{\frac{2\pi}{\beta m \omega^2}}

Z_1 = \frac{Z_p Z_q}{h} = \frac{k_B T}{\hbar \omega}

U = N k_B T, \quad P = \frac{N k_B T}{A} \quad (2D \text{ pressure})

k_B T_c = \hbar \omega \left(\frac{N}{g_s \zeta(d)} \right)^{1/d}

5. Diatomic molecule:

Z_1 = Z_{trans} \cdot Z_{rot} \cdot Z_{vib}, \quad S_{\text{total}} = S_{\text{trans}} + S_{\text{rot}} + S_{\text{vib}}

Z_{trans} = \frac{V}{\lambda^3}, \quad Z_{rot} = \frac{T}{2\Theta_r} \quad (\Theta_r = \frac{\hbar^2}{2Ik_B})

Z_{vib} = \frac{1}{2 \sinh(\Theta_v/2T)} \quad (\Theta_v = \frac{\hbar \omega}{k_B})

High T:

U = \frac{3}{2} N k_B T + N k_B T + N k_B T = \frac{7}{2} N k_B T

k_B T \ll \hbar \omega_{\text{vib}} \Rightarrow \text{vibrations inactive}

Spin & Magnetic Systems

6. Spin-1/2 in field B:

Z_1 = 2 \cosh(\beta \mu_B B)

\langle \mu \rangle = \mu_B \tanh(\beta \mu_B B), \quad M = N \langle \mu \rangle

\chi = \frac{N \mu_B^2}{k_B T} \quad (\text{Curie law})

7. N spins, total magnetization:

Z_N = (2 \cosh(\beta \mu_B B))^N

F = -N k_B T \ln(2 \cosh(\beta \mu_B B))

Quantum Statistics

9. Fermi gas at T = 0:

N = \int_0^{\epsilon_F} g(\epsilon) d\epsilon \Rightarrow \epsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}

U_0 = \int_0^{\epsilon_F} \epsilon g(\epsilon) d\epsilon = \frac{3}{5} N \epsilon_F

10. BEC critical temperature:

N = \int_0^\infty \frac{g(\epsilon)}{e^{\beta \epsilon} - 1} d\epsilon \quad \text{with } \mu = 0

\Rightarrow k_B T_c = \frac{2\pi \hbar^2}{m} \left(\frac{n}{\zeta(3/2)} \right)^{2/3}

11. Photon gas energy:

U = \int_0^\infty \frac{\hbar \omega \cdot g(\omega)}{e^{\beta \hbar \omega} - 1} d\omega \propto T^4

Grand Canonical Examples

12. Classical ideal gas:

\Xi = e^{zV/\lambda^3}, \quad \langle N \rangle = zV/\lambda^3,

PV = k_B T \ln \Xi = \langle N \rangle k_B T

Calculation:

\Xi = \sum_{N=0}^\infty z^N \frac{(Z_1)^N}{N!} = \sum_{N=0}^\infty \frac{(zZ_1)^N}{N!} = e^{zZ_1}

With Z_1 = V/\lambda^3: \Xi = e^{zV/\lambda^3}

\langle N \rangle = k_B T \frac{\partial \ln \Xi}{\partial \mu} = z \frac{\partial \ln \Xi}{\partial z} = zV/\lambda^3

\Phi = -k_B T \ln \Xi = -PV \Rightarrow PV = \langle N \rangle k_B T

13. Harmonic trap (Bose):

g(\epsilon) = \frac{\epsilon^2}{2(\hbar \omega)^3}, \quad T_c \propto N^{1/3} \hbar \omega

14. Two-level system entropy:

S = N k_B \left[\ln(1 + e^{-\beta \epsilon}) + \frac{\beta \epsilon}{1 + e^{\beta \epsilon}} \right]

Ising & Phase Transitions

16. Mean field Ising:

Near T_c:

m \approx \tanh(\beta z J m)

\approx \beta z J m - \frac{(\beta z J m)^3}{3}

\Rightarrow m^2 = 3 \frac{T_c - T}{T_c}

m \propto (T_c - T)^{1/2} \quad (\beta = 1/2)

Miscellaneous

17. Rubber band (F = aT(L - L_0)):

dU = T dS + F dL, \quad S = -\frac{a}{2} (L - L_0)^2 + S_0

18. Chemical equilibrium (A \rightleftharpoons B):

\mu_A = \mu_B \Rightarrow k_B T \ln(n_A \lambda_A^3) - \epsilon_A = k_B T \ln(n_B \lambda_B^3) - \epsilon_B

\frac{n_B}{n_A} = \left(\frac{\lambda_A}{\lambda_B} \right)^3 e^{-\beta(\epsilon_B - \epsilon_A)}

19. Effusion rate:

\Phi = \frac{1}{4} n \langle v \rangle = \frac{P}{\sqrt{2\pi m k_B T}}

20. Fluctuations:

\langle (\Delta E)^2 \rangle = k_B T^2 C_V, \quad \langle (\Delta N)^2 \rangle = k_B T \frac{\partial \langle N \rangle}{\partial \mu}

Mathematical Tools

Stirling & Gamma

\ln N! \approx N \ln N - N

\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx, \quad \Gamma(n+1) = n!, \quad \Gamma(1/2) = \sqrt{\pi}

Reflection formula: \Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin(\pi z)}

Duplication: \Gamma(z) \Gamma(z+1/2) = \frac{\sqrt{\pi}}{2^{2z-1}} \Gamma(2z)

Gaussian Integrals

\int_{-\infty}^\infty e^{-\beta p^2/2m} dp = \sqrt{\frac{2\pi m}{\beta}} = \frac{h}{\lambda}

\int_{-\infty}^\infty x^2 e^{-ax^2} dx = \frac{1}{2a} \sqrt{\frac{\pi}{a}}

\int_{-\infty}^\infty x^4 e^{-ax^2} dx = \frac{3}{4a^2} \sqrt{\frac{\pi}{a}}

\int_0^\infty x^n e^{-ax^2} dx = \frac{\Gamma(\frac{n+1}{2})}{2a^{(n+1)/2}}

\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}

Other Integrals

\int x e^{-ax} dx = -\frac{e^{-ax}}{a^2} (ax + 1) + C, \quad \int \frac{dx}{e^x - 1} = \ln(1 - e^{-x}) + C, \quad \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15} \quad (\text{Stefan-Boltzmann})

Series

Geometric: \sum_{n=0}^\infty x^n = \frac{1}{1-x} \quad (|x| < 1), \quad \sum_{n=0}^\infty n x^n = \frac{x}{(1-x)^2}

Taylor: e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}, \ln(1+x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}, \ln(1-x) \approx -x - \frac{x^2}{2}, (1+x)^n \approx 1 + nx + \frac{n(n-1)}{2} x^2, \sin x \approx x - \frac{x^3}{6}, \cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}, \frac{1}{1-x} \approx 1 + x + x^2, \frac{1}{1+x} \approx 1 - x + x^2

Hyperbolic Functions

\cosh x = \frac{e^x + e^{-x}}{2}, \sinh x = \frac{e^x - e^{-x}}{2}, \tanh x \approx x - \frac{x^3}{3}

(small x), \tanh x \rightarrow 1 \text{ (large x)}, \text{sech}^2 x = 1 - \tanh^2 x

Jacobians for Coordinate Changes

Polar (2D): x = r \cos \theta, y = r \sin \theta \Rightarrow \mathcal{J} = r

Spherical (3D):

x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta

\Rightarrow \mathcal{J} = r^2 \sin \theta

General:

\int_V f(\mathbf{r}) d^3 r = \int_0^\infty \int_0^\pi \int_0^{2\pi} f(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi

Spherically symmetric (f = f(r)):

\int_V f(r) d^3 r = 4\pi \int_0^\infty f(r) r^2 dr

Ultra-relativistic partition function example:

$\epsilon = pc,$

$$Z_1 = \frac{V}{h^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} e^{-\beta pc} p^2 \sin \theta \, dp \, d\theta \, d\phi$$

$\int_0^\pi \sin \theta \, d\theta = 2, \int_0^{2\pi} d\phi = 2\pi \Rightarrow 4\pi$

$$\boxed{Z_1} = \frac{4\pi V}{h^3} \int_0^\infty p^2 e^{-\beta pc} dp$$
$$= \frac{8\pi V (k_B T)^3}{h^3 c^3}$$

using $\int_0^\infty p^2 e^{-\beta pc} dp = \frac{2}{(\beta c)^3},$

$U = 3Nk_B T, PV = U/3$

Combinatorics

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \text{ Stars \& bars: } n \text{ balls in } k \text{ cells:}$$
$$\binom{n+k-1}{k-1}$$

$\binom{n}{k} = \binom{n}{n-k}$ (symmetry), $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$ (Pascal), $\sum_{k=0}^n \binom{n}{k} = 2^n,$

$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}$

N-dim Sphere

$$V_N(R) = \frac{\pi^{N/2}}{\Gamma(N/2+1)} R^N, \quad \Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)} \text{ (surface area)}$$

Polylogarithm

$$\text{Li}_n(z) = \sum_{k=1}^\infty \frac{z^k}{k^n}, \quad \zeta(n) = \text{Li}_n(1), \quad \zeta(3/2) \approx 2.612, \quad \zeta(5/2) \approx 1.341$$

Dimensional Analysis

Quantity	Check Unit	Dim (M, L, T, Θ)
Energy (U, F, μ)	k _B T, PV	ML ² T ⁻²
Pressure (P)	U/V	ML ⁻¹ T ⁻²
Entropy (S, C)	k _B , U/T	ML ² T ⁻² Θ ⁻¹
DOS g(ε)	N/ε	M ⁻¹ L ⁻² T ²
Action (h)	x · p	ML ² T ⁻¹
Length (λ _h)	ħ/√mkT	L
Density (n _Q)	1/λ ³	L ⁻³

Checks: [ε_F] = [ħ²n^{2/3}/m] = energy ✓

Characteristic Temperatures

System	Char. Temp.
Fermi gas	T _F = ε _F /k _B
BEC	T _c ∝ n ^{2/3} /m
Debye solid	Θ _D = ħω _D /k _B
Rotation	Θ _r = ħ ² /(2Ik _B)
Vibration	Θ _v = ħω/k _B

Heat Capacity Summary

System	Low T	High T
Fermi gas	C _V ∝ T	$\frac{3}{2}Nk_B$
BEC (T < T _c)	C _V ∝ T ^{3/2}	—
Debye solid	C _V ∝ T ³	3Nk _B
Photon gas	C _V ∝ T ³	—
HO (quantum)	∝ e ^{-Θ_v/T}	Nk _B

Exam Specials
Dimensional Scaling (Virial)

For $H = \sum A p^s + \sum B q^n$ in d dims:

$$U = Nk_B T \left(\frac{d}{s} + \frac{d}{n} \right)$$

Common Potentials (d = 3):

- Harmonic** (n = 2): 3/2 + 3/2 = 3 → 3Nk_BT
- Linear Trap** (n = 1): 3/2 + 3/1 = 4.5 → 4.5Nk_BT
- Box** (n → ∞): 3/2 + 0 = 1.5 → 1.5Nk_BT
- Quartic** (n = 4): 3/2 + 3/4 = 2.25 → 2.25Nk_BT

Generalized Occupancy

For particles with restricted/custom occupancy $n \in \{0, 1, \dots, k\}$: Do **not** use FD/BE. Build Ξ for single state:

$$\Xi_{state} = \sum_n (ze^{-\beta \epsilon})^n$$
$$\langle n \rangle = z \frac{\partial \ln \Xi}{\partial z}$$

”Pairons” ($n \in \{0, 1, 2\}$): Let $x = ze^{-\beta \epsilon}$

$$\Xi = 1 + x + x^2 \implies \langle n \rangle = \frac{x + 2x^2}{1 + x + x^2}$$

Spin-1 Ising ($S \in \{-1, 0, 1\}$)
Partition (Free): $Z_1 = 1 + 2 \cosh(\beta h_{eff})$ **Mean Field:** $h_{eff} = zJm + h$

$$m = \frac{2 \sinh(\beta h_{eff})}{1 + 2 \cosh(\beta h_{eff})}$$

Critical Temp ($\Delta = 0$): $k_B T_c = \frac{2}{3} zJ$

2D Relativistic (Graphene)
Dispersion $\epsilon = \hbar v|k|$ in $d = 2$. **DOS:** $g(\epsilon) = \frac{g_s}{2\pi(\hbar v)^2} |\epsilon| \propto \epsilon$ **Thermo:** $U \propto T^3, C_V \propto T^2$ **Particle-Hole:** If symmetric, $\mu(T = 0) = 0$.

Fermi Fluctuations

$$\sigma_n^2 = \langle n^2 \rangle - \langle n \rangle^2 = \langle n \rangle (1 - \langle n \rangle)$$

Max fluctuation at $\epsilon = \mu$ (where $\langle n \rangle = 0.5$) → $\sigma^2 = 0.25$.

Potentials
Logarithmic $V(r) = \phi_0 \ln(r/R)$: Spatial integral $\int_0^R e^{-\beta \phi_0 \ln(r/R)} r^2 dr$ converges only if $3 - \beta \phi_0 > 0$. **Spherical Linear** $V(r) = ar$: $\int_0^\infty e^{-\beta ar} r^2 dr \propto \beta^{-3} \implies Z_{conf} \propto T^3$.

Exam Solutions & Recipes
1. Generalized Statistics (”Pairons” etc.)
Problem: Particles where a single quantum state can hold $n \in \{0, 1, \dots, k\}$ particles with energy $n\epsilon$. (e.g., ”Pairons” $k = 2$). **Solution Method:**
1. Grand Partition (Single State):

$$\xi(\epsilon) = \sum_{n=0}^k (ze^{-\beta \epsilon})^n = \sum_{n=0}^k X^n$$

- where $X \equiv e^{\beta(\mu - \epsilon)}$.
- 2. Geometric Series:** If $k \rightarrow \infty$ (Bosons): $\xi = \frac{1}{1-X}$. If $k = 1$ (Fermions): $\xi = 1 + X$. If $k = 2$ (Pairons): $\xi = 1 + X + X^2$.
- 3. Average Occupancy:**

$$\langle n(\epsilon) \rangle = z \frac{\partial \ln \xi}{\partial z} = X \frac{\partial \ln \xi}{\partial X}$$

For Pairons ($k = 2$):

$$\langle n \rangle = \frac{X + 2X^2}{1 + X + X^2}$$

- 4. T = 0 Limit (Fermi Energy):**
Max occupancy is k . Fill states up to ϵ_F .

$$N = \int_0^{\epsilon_F} g(\epsilon) \cdot k d\epsilon$$

2. Spin-1 Ising Model (Mean Field)
Problem: Spins $s_i \in \{-1, 0, +1\}$. Hamiltonian $H = -J \sum_{\langle ij \rangle} s_i s_j - h \sum s_i$. **Solution Method:**

- 1. Effective Field:** Replace neighbors with mean m .

$$h_{eff} = zJm + h$$

- 2. Single Particle Z:**

$$Z_1 = \sum_{s=-1,0,1} e^{\beta h_{eff} s} = e^{-\beta h_{eff}} + e^0 + e^{\beta h_{eff}}$$

$$\boxed{Z_1 = 1 + 2 \cosh(\beta h_{eff})}$$

- 3. Magnetization Equation:**

$$m = \langle s \rangle = \frac{1}{Z_1} \sum_s s e^{\beta h_{eff} s} = \frac{2 \sinh(\beta h_{eff})}{1 + 2 \cosh(\beta h_{eff})}$$

- 4. Critical Temperature (T_c):** Set $h = 0$. Expand for small m ($\sinh x \approx x, \cosh x \approx 1$).

$$m \approx \frac{2(\beta zJm)}{1 + 2} = \frac{2}{3} \beta zJm$$

Divide by $m \neq 0$: $1 = \frac{2}{3} \frac{zJ}{k_B T_c} \Rightarrow \boxed{T_c = \frac{2zJ}{3k_B}}$

3. Dimensional Scaling (Virial)
Problem: Find $U(T)$ or C_V for non-quadratic potentials $H \propto |q|^n$ or $|p|^s$ without full integration. **Recipe:** $Z \propto (\text{Length})^d \times (\text{Momentum})^d$.

1. Momentum scaling: $E_K \propto p^s \rightarrow p \propto T^{1/s}$. Integral $\int e^{-\beta p^s} d^d p \propto T^{d/s}$.

2. Position scaling: $V(q) \propto q^n \rightarrow q \propto T^{1/n}$. Integral $\int e^{-\beta q^n} d^d q \propto T^{d/n}$.

3. Total Partition Function:

$$Z(T) \propto T^{\frac{d}{s} + \frac{d}{n}}$$

- 4. Energy & Heat Capacity:**

$$U = \left(\frac{d}{s} + \frac{d}{n} \right) Nk_B T, \quad C_V = \frac{U}{T}$$

Specific Cases (3D, s = 2):

- Harmonic ($n = 2$): $U = (\frac{3}{2} + \frac{3}{2})Nk_B T = 3Nk_B T$.
- Linear Trap ($V = ar, n = 1$): $U = (\frac{3}{2} + \frac{3}{1})Nk_B T = 4.5Nk_B T$.
- Box ($n = \infty$): $U = (\frac{3}{2} + 0)Nk_B T = 1.5Nk_B T$.

4. 2D Relativistic Gas (Graphene)
Problem: 2D fermions with dispersion $\epsilon = \hbar vk$. **Solution Method:**

1. Density of States: Phase space $N(\epsilon) = g_s \frac{A}{(2\pi)^2} (\pi k^2)$. Substitute $k = \epsilon/\hbar v$: $N(\epsilon) \propto \epsilon^2$.

$$g(\epsilon) = \frac{dN}{d\epsilon} \propto \epsilon \quad (\text{Linear!})$$

Exact: $g(\epsilon) = \frac{g_s A}{2\pi(\hbar v)^2} |\epsilon|$

- 2. Energy (T → 0):** $U \propto \int \epsilon g(\epsilon) f(\epsilon) d\epsilon$. At low T , $\propto T^{d+1}$ for massless. Since $g(\epsilon) \sim \epsilon$, integrand is $\epsilon^2 e^{-\beta \epsilon}$.

$$U \propto T^3, \quad C_V \propto T^2$$

(Compare to 3D Phonons: $U \propto T^4, C_V \propto T^3$).

5. Adsorption on Heterogeneous Surface
Problem: N particles absorb on M sites, where site energies ϵ are distributed via $P(\epsilon) = Ae^{\epsilon/\Delta}$ for $-2\Delta < \epsilon < 0$. **Solution Method:**

1. Single Site Partition: $Z_{site}(\epsilon) = 1 + e^{-\beta(\epsilon - \mu)}$ (Fermion-like, site can be empty or full).

2. Average Partition/Occupancy: Do *not* integrate Z . Integrate the physical quantity (occupancy).

$$\langle N \rangle = M \int P(\epsilon) \frac{1}{e^{\beta(\epsilon - \mu)} + 1} d\epsilon$$

- 3. Approximation:** If $T \rightarrow 0$, Fermi function is step at μ .

$$\langle N \rangle \approx M \int_{-2\Delta}^{\mu} P(\epsilon) d\epsilon$$

T_c with Gap Δ: Condition $\mu(T_c) = \epsilon_{min} = -\Delta$. Total density $n = n_{s=-1} + n_{s=+1}$. Using $g(\epsilon) \propto \sqrt{\epsilon \pm \Delta}$:

$$n = \int_{-\Delta}^{\infty} \frac{g_{-1}(\epsilon) d\epsilon}{e^{\beta(\epsilon + \Delta)} - 1} + \int_{\Delta}^{\infty} \frac{g_{+1}(\epsilon) d\epsilon}{e^{\beta(\epsilon + \Delta)} - 1}$$

Sub $x = \beta(\epsilon + \Delta)$ for spin -1, $x = \beta(\epsilon - \Delta)$ for spin +1:

$$n = \left(\frac{mk_B T_c}{2\pi \hbar^2} \right)^{3/2} \left[\zeta\left(\frac{3}{2}\right) + \text{Li}_{3/2}\left(e^{-\frac{2\Delta}{k_B T_c}}\right) \right] \quad (2)$$

(Implicit equation for T_c . First term: ground branch saturation. Second term: upper branch.)

Units & Constants:

- Energy:** 1 eV = 1.602 × 10⁻¹⁹ J = 1.602 × 10⁻¹² erg.
- CGS Connection:** 1 J = 10⁷ erg = 10⁷ (dyn · cm).
- Force:** 1 dyn = 10⁻⁵ N $\implies$ 1 N = 10⁵ dyn.
- Temp:** $k_B = 1.38 \times 10^{-23}$ J/K = 8.617 × 10⁻⁵ eV/K.